



Semester: Spring 2025

Max Marks: 9

Track: Track B – Software-Based AI System

Mapped CLO: CLO-3

Instructor: Dr. Shafiq Ur Rehman Khan

Track B – Software-Based AI System
Milestone 3 – Final Product and Complete System

Course: CSC-361 Machine Learning	Semester: Spring 2025
Track: Track B – Software-Based AI System	Due Date: 30 May 2025
Instructor: Dr. Shafiq Ur Rehman Khan	Max Marks: 9 / 20
Mapped CLO: CLO-3	

Mapped Course Learning Outcome (CLO)

- CLO-3: Analyze machine learning model performance using appropriate preprocessing techniques and evaluation metrics.

Milestone Objective

Students will deliver a fully functioning AI system with a complete ML pipeline, live demonstration, system documentation, and viva. This milestone evaluates the rigor of model performance analysis including preprocessing, evaluation metrics, and result visualization within a real-world software system.

Required Deliverables

- Final Written Report / System Documentation (PDF, 5–8 pages)
- Live Demonstration + Viva (10–12 minutes)
- Final cleaned GitHub repository with run instructions

Submission Requirements

- Final system documentation including updated architecture diagram
- Complete ML model evaluation with metrics and visualizations
- Testing results (unit tests or integration tests)
- Comparison of at least two model configurations or algorithms
- User guide explaining how to use the system
- Discussion of challenges, limitations, and possible improvements

Evaluation Rubric — Total Marks: 9 / 20



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Group Name / No.:

Submission Date:

Criterion	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Needs Work (25%)	Marks	CLO
Final ML Model Quality	Well-tuned model with strong performance; results exceed baseline significantly	Good model; moderate improvement over baseline	Model functional but performance is weak	Non-functional or untrained model	2.0	CLO-3
Model Comparison & Tuning	Two+ configurations/ algorithms compared; tuning documented with results	Comparison present; tuning partially documented	Minimal comparison; no tuning	No comparison or tuning	1.5	CLO-3
Evaluation & Visualization	Multiple metrics, graphs (confusion matrix, ROC/PR curve, loss plot), and deep analysis	Metrics and graphs present with moderate analysis	Some metrics reported; limited visualization	No evaluation or visualization	2.0	CLO-3
System Completeness & Documentation	Complete system with user guide, architecture diagram, and full documentation	System mostly complete; documentation has gaps	Partial system; limited documentation	Incomplete system; no documentation	1.0	CLO-3
Live Demo & System Functionality	System runs end-to-end live; all features demonstrated without errors	Demo mostly successful; minor issues	Demo partially functional	Demo failed or not attempted	1.5	CLO-3
Viva (Individual Contribution)	Each member confidently explains their contribution and answers all questions	Most members answer adequately	Some members unable to explain their part	Members cannot explain the project	0.75	CLO-3
Code Quality & Reproducibility	Clean repo; code runs with one command; reproducible results; proper commit history	Code runs with minor setup; mostly clean	Code runs with significant effort; messy	Code non-functional or missing	0.25	CLO-3

Marks Distribution Summary

Criterion	Max Marks	CLO
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Final ML Model Quality	2.0	CLO-3
Model Comparison & Tuning	1.5	CLO-3
Evaluation & Visualization	2.0	CLO-3
System Completeness & Documentation	1.0	CLO-3
Live Demo & System Functionality	1.5	CLO-3
Viva (Individual Contribution)	0.75	CLO-3
Code Quality & Reproducibility	0.25	CLO-3
Total	9.00	

Important Notes: All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.